

# PARTEEK GARG

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[LinkedIn](#) | [GitHub](#) | [LeetCode](#) | [Portfolio](#)

## Summary

B.Tech Computer Science (Data Science) student who builds and ships production-grade full-stack systems rather than course projects. Independently designed and developed three complete applications — a RAG-based fact-verification platform, a bioinformatics simulator, and a client-side academic management system — covering authentication, database design, API architecture, and applied AI. Oracle Cloud certified in AI and infrastructure foundations; active problem solver on LeetCode.

## Skills

### Programming Languages Frontend

Python · Java · JavaScript · TypeScript · C  
React.js · HTML5 · CSS3 · Tailwind CSS · Framer  
Motion

### Backend & Tools Databases

Node.js · Express · Git · GitHub · VS Code · Vite  
Supabase (PostgreSQL) · Drizzle ORM ·  
localStorage

### AI / Data Science

Groq LLM · RAG Architecture · Data Science  
(learning)

### Domains

Full Stack Web · Bioinformatics · Academic Tools

## Education and Training

**B.Tech: Computer Science (Data Science)**  
SVKM's NMIMS

2025-Present  
Chandigarh

## Certifications

|                                                                 |          |
|-----------------------------------------------------------------|----------|
| Introduction to Generative AI — Simplilearn                     | Feb 2026 |
| Oracle Cloud Infrastructure 2025 Foundations Associate — Oracle | Feb 2026 |
| OCI AI Foundations Associate — Oracle                           | Feb 2026 |
| Machine Learning for Data Science Projects — IBM                | May 2026 |
| Data Analytics Essentials — Cisco Networking Academy            | May 2026 |

## Projects

### Campus Care

[Github](#)

*Vanilla TypeScript · Auth0 · jsPDF · Playwright · localStorage*

- Built a fully client-side academic platform with attendance tracking, CGPA/SGPA calculation, and what-if simulation — developed without any UI framework using pure TypeScript DOM manipulation
- Implemented Auth0 authentication with per-user localStorage scoping and an automatic silent data migration engine supporting legacy multi-semester data structures
- Added client-side PDF report generation (jsPDF) and Playwright E2E test coverage; developed following the complete Design Thinking framework as part of the TII course

### DNA Encoding

[Github](#)

*React · TypeScript · Node.js · Express · Framer Motion · Drizzle ORM · PostgreSQL*

- Built a full-stack interactive simulator converting text to DNA nucleotide sequences via a deterministic binary-chunking pipeline with configurable base mappings and live validation

- Rendered a live animated SVG double helix using Framer Motion responding to real-time input changes, with nucleotide composition stats and an educational insights panel
- Expanded from a single HTML file university assignment into a production-grade application with PostgreSQL persistence, REST API, TanStack React Query, and in-memory fallback

## Veralon

[Github](#)

*React · TypeScript · Node.js · Express · Groq LLM · Supabase · Auth0 · Drizzle ORM*

- Built a RAG-based credibility verification platform evaluating claims (text, URL, PDF, image) against up to 30 deduplicated sources via a deterministic 13-step verification pipeline
- Engineered an 8-component 100-point scoring engine using pure arithmetic — zero LLM involvement in final scoring math — ensuring fully reproducible, auditable results with SHA-256 audit trails
- Implemented Union-Find evidence clustering, Row-Level Security on all Supabase tables, Auth0 authentication, OWASP-aligned rate limiting, and Zod input validation throughout

## Lumiere — Hack-O-Mania 2.0 (Team Lead, Team of 5)

[Github](#)

*Next.js · React · FastAPI · PostgreSQL · Python · TailwindCSS · AI/ML*

- Led architecture, product direction, and system design for an AI-powered patient identity resolution system— built real-time duplicate medical record detection using hybrid exact, fuzzy, and ML-inspired matching logic
- Designed a confidence-based scoring workflow with a multi-stage human-in-the-loop review pipeline and privacy-preserving AI processing, architected around a doctor-centric dashboard structure
- Coordinated a cross-functional team of 5 across frontend, backend, and AI integration — led research, presentation flow, and full project delivery under hackathon constraints

## GreenTrace — Environmental Intelligence Platform

[Github](#)

*Next.js, React, TypeScript, Tailwind CSS, Node.js, PostgreSQL, Drizzle ORM*

- Built a full-stack sustainability intelligence platform that analyzes personal carbon footprints, visualizes environmental impact, and delivers personalized recommendations through an interactive analytics dashboard.
- Designed and implemented the Carbon Twin Simulator, enabling users to model lifestyle changes and instantly project their impact on annual CO<sub>2</sub> emissions before adopting new habits.
- Developed premium product features including Carbon Signature Cards, progress tracking, sustainability challenges, personalized recommendation engine, and responsive dashboards with a production-grade UI/UX.